Investigation of the Attributes of the Sport of Boxing and the Changes Over the Years

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Introduction

The sport of boxing has evolved over the centuries, and so has the average boxer. This is due to the fact that rules have changed, and that the people have changed. I want to see what has stayed consistent and what hasn't, to observe what styles and attributes are considered attractive for a boxer. I will also be looking at other factors that could determine change in the sport, like the emergence of figures who had an impact on the martial art.

I hypothesize that the most effective boxing style will either be an out-boxer or a boxer-puncher, due to the fact that out-boxers are inherently the safest fighters, guaranteeing to succumb to less cumulative damage over time. I also believe the other contender for most effective fight style would be the boxer-puncher, as they do not have any natural counters, bearing this in mind I think that out-boxers and boxer-punchers will hold the belt for the longest time simply due to their longevity.

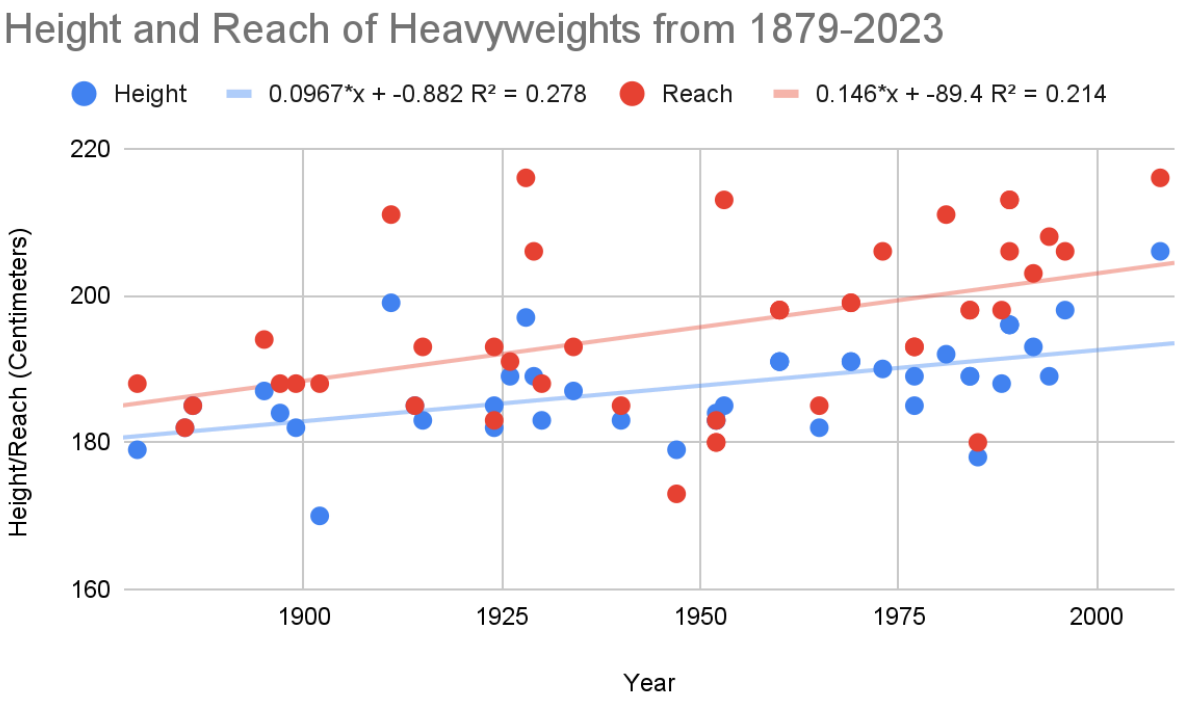
Information will be gathered from trusted sources used by boxing officials, like CompuBox and BoxRec. The population I will be taking is of the lineal champions of the heavyweight and light welterweight division.

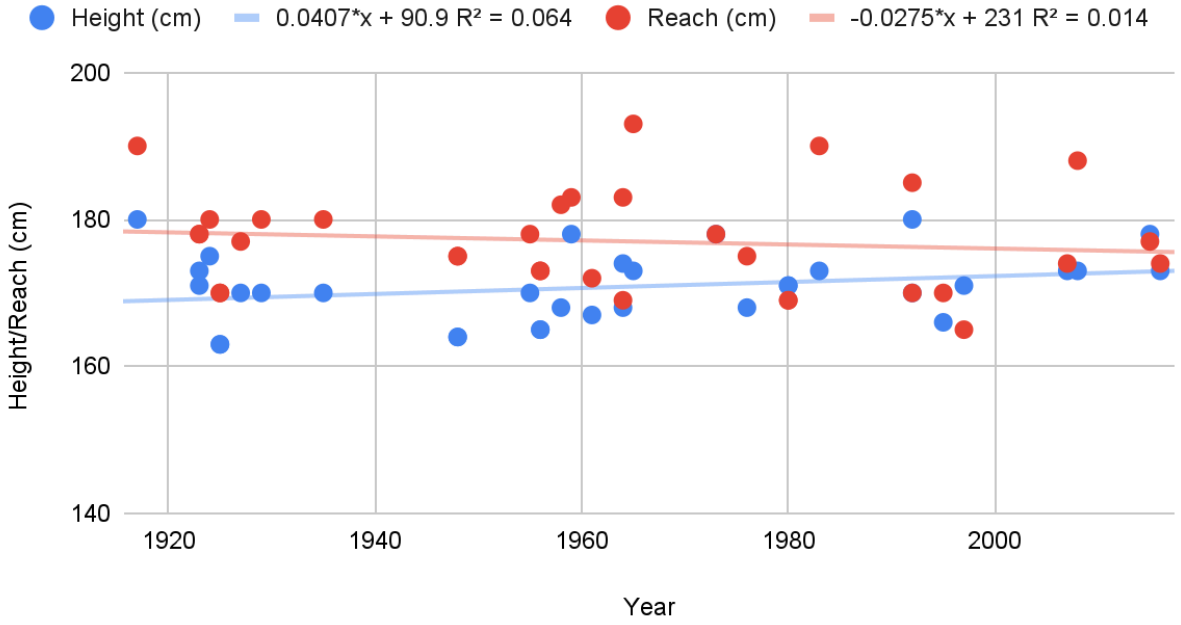
The main question I will be answering is how has the average champion changed over the years? (Height, Ape Index, Stance etc...) I will also be asking other questions like:

- how do the physical attributes of the boxer shape their style? For example, does Ape Index help with power?

- -What has remained the same over the years? (Height, Ape Index, Stance, etc...)
- How do weight classes affect the way people fight?

DATA ANALYSIS





These two graphs can be considered continuous, as a champion could lose the belt during the middle of the year, making it so the height and reach of the champion could change within a decimal of any given year, despite it being unconventional to refer to years in fractions. Height and Reach of champions are weakly correlated with time past. This is because the rate at which humans grow over the years is very slow, so with only around 100-120 years to account for, the average height and reach would have barely budged. For example, the average height 100 years ago was 171 cm. Now, the average height is 175 cm. The welterweights on the other hand had no correlation of height and reach over the years. They have stayed the same size all of this time. This makes a lot of sense, since if Welterweights were to grow too tall, they would also weigh too much to be in that division, and so they will simply move onto the next weight class, which isn't the case with heavyweights due to that being the maximum weight class; they don't have any weight class to move to. The graph may look biased since its base starts at 160/145 for heavyweight and light welterweight accordingly, but this is just so it accurately reflects the difference in height between fighters. Those few inches mean a lot. Also, if it were to start at zero the points would clump up too much and the graph would be unusable. The coefficient of determination shows us that time has little effect on height for heavyweights and none for light welterweights regarding height.

Heavyweight Height Over the Years

Correlation Coefficient	Coefficient of Determination
r=0.527	r^2=0.278

Welterweight Height Over the Years

Correlation Coefficient	Coefficient of Determination
r=0.253	r^2=0.064

Heavyweight Reach Over the Years

Correlation Coefficient	Coefficient of Determination
r=0.463	r^2=0.214

Welterweight Reach Over the Years

Correlation Coefficient	Coefficient of Determination

r=0.1	r^2=0.01

LINEAR REGRESSION HEAVYWEIGHT HEIGHT

Σx	Σ of Years	85763
Σy	Σ Height	7886
Mean of Years		1949.16
Mean Height		187.636

$$Y = 0.8818 + 0.09672X$$

Extrapolation

What will the heavyweight champ's height be in 2050?

Y = 0.8818 + 0.09672(2050)

Y = 199.1578

His height will be approximately 199.1578 centimeters

Interpolation

What was the heavyweight champ's height in 1970?

Y = 0.8818 + 0.09672(1970)

Y = 191.4202

His height should be approximately 191.4202 centimeters

We can compare this to the actual height of the champ of 1970, Joe Frazier. His height was 182 cm. The inaccuracy should have been obvious, as the correlation coefficient between height over time isn't that high.

LINEAR REGRESSION WELTERWEIGHT HEIGHT

Σx	Σ of Years	64776
Σy	Σ Height	5586
Mean of Years		1962.91
Mean Height		170.788

$$Y = 90.9427 + 0.04068X$$

Extrapolation

What will the light welterweight champ's height be in 2050?
Y = 90.9427 + 0.04068(2050)
Y = 174.3367
His height will be approximately 174.3367 centimeters

Interpolation

What was the light welterweight champ's height in 1970?
Y = 90.9427 + 0.04068(1970)
Y = 171.0823
His height should be approximately 171.0823 centimeters

We can compare this to the actual height of the champ of 1970, Nicolino Locche. His height was 168 cm. This was extremely accurate, and I think this is due to the fact that there has been little to no growth in the champions of the light welterweight division, for reasons explained earlier.

LINEAR REGRESSION HEAVYWEIGHT REACH

Σx	Σ of Years	85763
Σy	Σ Reach	8605
Mean of Years		1949.16
Mean Reach		195.568

$$-89.4447 + 0.1462X$$

Extrapolation

What will the heavyweight champ's reach be in 2050?
Y = -89.4447 + 0.1462X(2050)
Y = 210.2653
His reach will be approximately 210.2653 centimeters.

Interpolation

What was the heavyweight champ's reach in 1970?
Y = -89.4447 + 0.1462(1970)
Y = 198.5693
His reach should be approximately 198.5693 centimeters

We can compare this to the actual reach of the champ of 1970, Joe Frazier. His reach was 185 cm. This was also inaccurate, for the same reasons mentioned above for the height regression line.

LINEAR REGRESSION WELTERWEIGHT REACH

Σx	Σ of Years	210790
Σy	Σ Reach	5843
Mean of Years		1962.91
Mean Reach		177.061

231.1195 - 0.02754X

Extrapolation

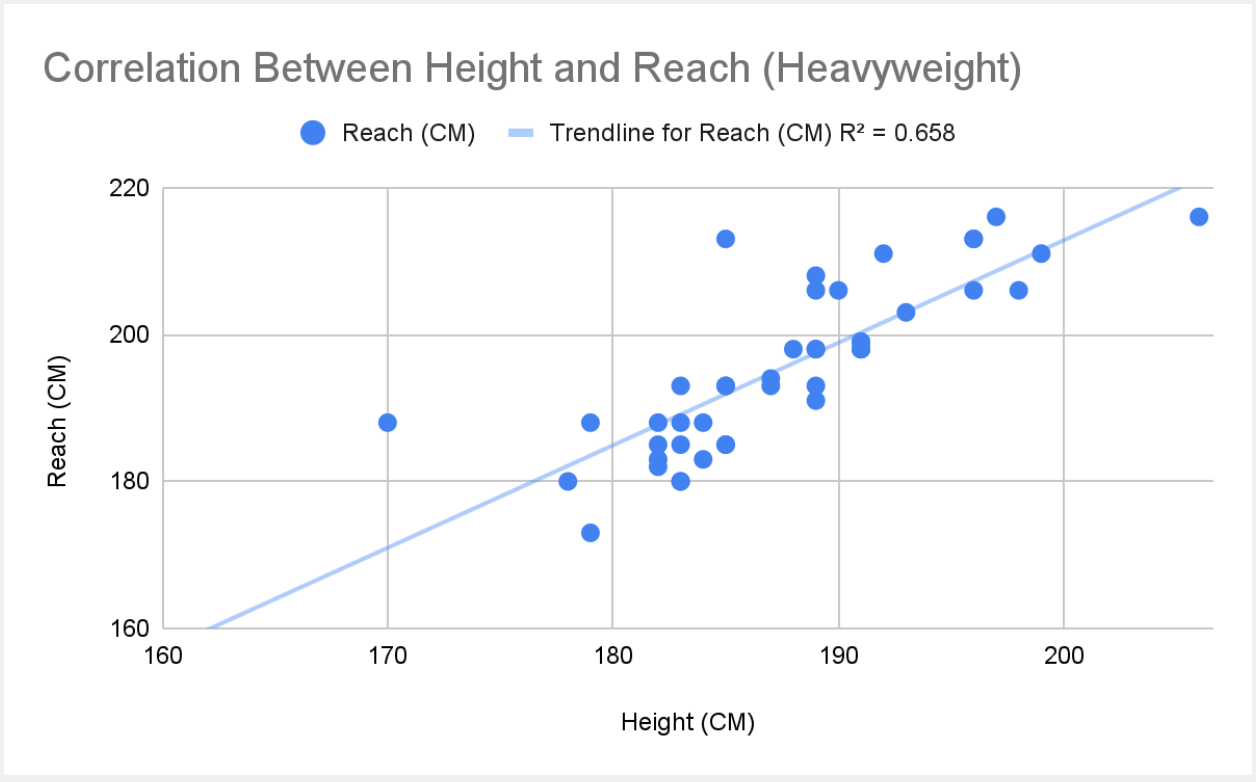
What will the light welterweight champ's reach be in 2050?
Y = 231.1195 - 0.02754(2050)
Y = 174.6625
His reach will be approximately 174.6625 centimeters.

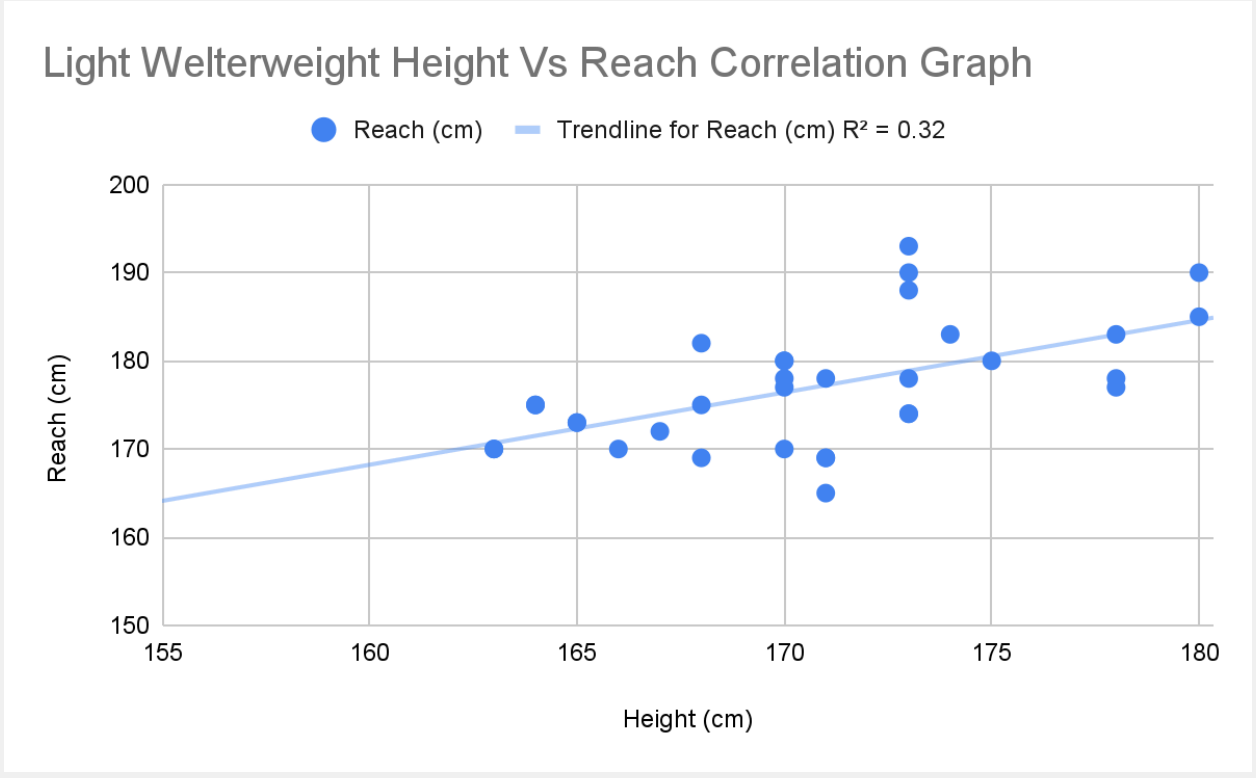
Interpolation

What was the light welterweight champ's reach in 1970?
Y = 231.1195 - 0.02754(1970)
Y = 176.8657
His reach should be approximately 176.8657 centimeters

We can compare this to the actual reach of the champ of 1970, Nicolino Locche. His reach was 182 cm. This was slightly inaccurate, but that is because Locche has an unusually high ape index, if he were to have the average ape index of 7 he would be extremely close to this estimate, however his ape index is double.

Correlation of Height and Reach





These two graphs can also be considered continuous, since it is possible to have a fraction of a centimeter. Height and Reach of champions are highly correlated. This is obviously due to the fact that a normal humans arms grows proportionally to their height.

However, what confuses me is why the correlation coefficient of the light welterweights is so much weaker than the heavyweights. Why are light welterweights much less proportional than the heavyweights? This could just be circumstantial, as the population size is small with a measly 33 subjects.

Heavyweight Height and Reach

Correlation Coefficient	Coefficient of Determination
r=0.811	r^2=0.658

Welterweight Height and Reach

Correlation Coefficient	Coefficient of Determination
r=0.566	r^2=0.32

MEASURES OF CENTRAL TENDENCY/SPREAD: *HEAVYWEIGHT HEIGHT (CM)*				
Mean	Median	Mode	Standard Deviation	Variance
187.636	187.5	189	6.4705307815543	41.867768595041

MEASURES OF CENTRAL TENDENCY/SPREAD: LIGHT *WELTERWEIGHT HEIGHT (CM)*				
Mean	Median	Mode	Standard Deviation	Variance
170.788	171	173	4.6953076238	22.04591368227

MEASURES OF CENTRAL TENDENCY/SPREAD: *HEAVYWEIGHT REACH (CM)*				
Mean	Median	Mode	Standard Deviation	Variance
195.568	193.5	198	11.132258055825	123.92716942149

MEASURES OF CENTRAL TENDENCY/SPREAD: LIGHT *WELTERWEIGHT REACH (CM)*				
Mean	Median	Mode	Standard Deviation	Variance
177.061	177	178	6.799908172015	46.2387511478

Heavyweight Ape Index

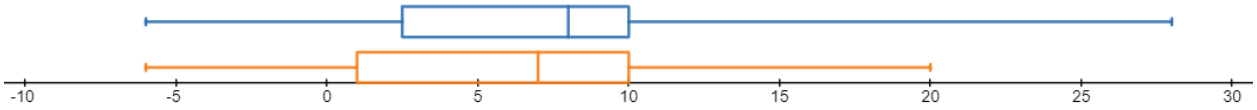
MEASURES OF CENTRAL TENDENCY/SPREAD: *HEAVYWEIGHT APE INDEX (CM)*

Mean	Median	Mode	Standard Deviation	Variance
7.932	8	10	6.998	48.973

Welterweight Ape Index

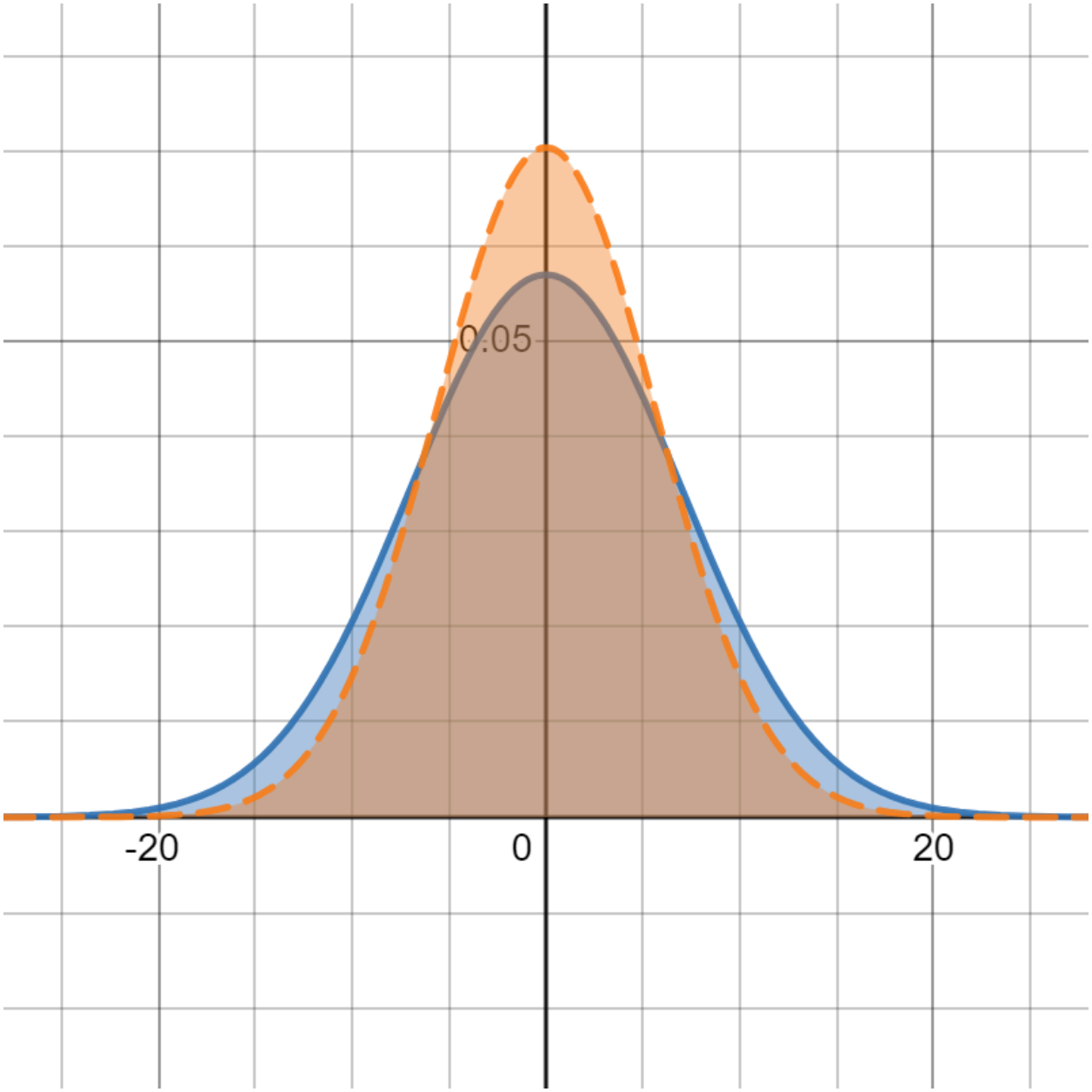
MEASURES OF CENTRAL TENDENCY/SPREAD: *WELTERWEIGHT APE INDEX (CM)*				
Mean	Median	Mode	Standard Deviation	Variance
6.272	7	7	5.669	32.138

Heavyweight
Welterweight



The interquartile range of the heavyweight ape index (Q3-Q1) equals 10-2.5, which is 7.5. For the welterweights, the interquartile range is 10-1, which is 9. These two box and whisker plots are extremely similar, which shows that there is little to no correlation between height and ape indexes, which would make sense because ape index subtracts height in the equation anyways, and every proportionate human should have an ape index of 0. The whiskers show that there are some extreme outliers on either end of the box, with Sonny Liston having an Ape Index of 28 and Rocky Marciano having an Ape Index of -6.

Having a standard deviation of 6.998 is extremely high, and so is a deviation of 5.669, which shows that there is a high amount of dispersion between the data from the mean value, which is 7.932 and 6.272 for heavyweight and light welterweight respectively.
These dispersions can be attributed to the massive exceptions mentioned above. These deviations do not explain any correlation between any physical attributes, it may suggest that ape indexes are random.



*Zoomed In Significantly

Heavyweight

Light Welterweight

Z-SCORE HEAVYWEIGHTS

Name	Champion	Ape Index	Z-Score
John L Sullivan	0	9	0.15262
James J Corbett	1	0	-1.13347
Bob Fitzsimmons	2	0	-1.13347
James J Jeffries	3	7	-0.13318
Marvin Hart	4	6	-0.27608
Tommy Burns	5	18	1.4387
Jack Johnson	6	4	-0.56187
Jess Willard	7	12	0.58131
Jack Dempsey	8	0	-1.13347
Gene Tunney	9	10	0.29551
Max Schmeling	10	8	0.0097171
Jack Sharkey	11	1	-0.99057
Primo Carnera	12	19	1.58159
Max Baer	13	17	1.2958
Jim Braddock	14	2	-0.84767
Joe Louis	15	6	-0.27608
Ezzard Charles	16	2	-0.84767
Jersey Joe Walcott	17	5	-0.41898
Rocky Marciano	18	-6	-1.99085
Floyd Patterson	19	-3	-1.56216
Ingemar Johansson	20	-1	-1.27636
Floyd	21	-3	1.56216

Patterson			
Sonny Liston	22	28	2.86768
Muhammad Ali	23	7	-0.13318
Joe Frazier	24	3	-0.70477
George Foreman	25	8	0.0097171
Muhammad Ali	26	7	-0.13318
Leon Spinks	27	8	0.0097171
Muhammad Ali	28	7	-0.13318
Larry Holmes	29	16	1.1529
Michael Spinks	30	4	-0.56187
Mike Tyson	31	2	-0.84767
James Douglas	32	19	1.58159
Evander Holyfield	33	9	0.15262
Riddick Bowe	34	10	0.29551
Evander Holyfield	35	9	0.15262
Michael Moorer	36	10	0.29551
George Foreman	37	8	0.0097171
Shannon Briggs	38	10	0.29551
Lennox Lewis	39	17	1.2958
Hasim Rahman	40	19	1.58159
Lennox	41	17	1.2958

Lewis			
Wladimir Klitschko	42	8	0.0097171
Tyson Fury	43	10	0.29551

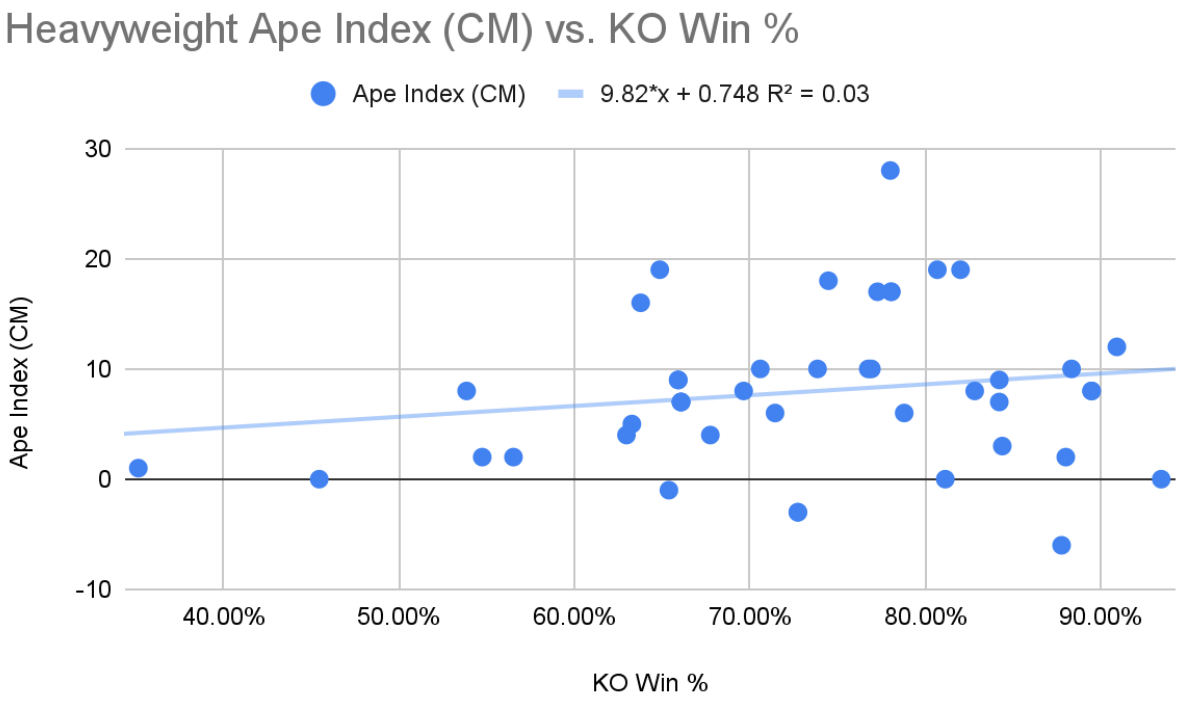
Z-SCORE LIGHT WELTERWEIGHTS

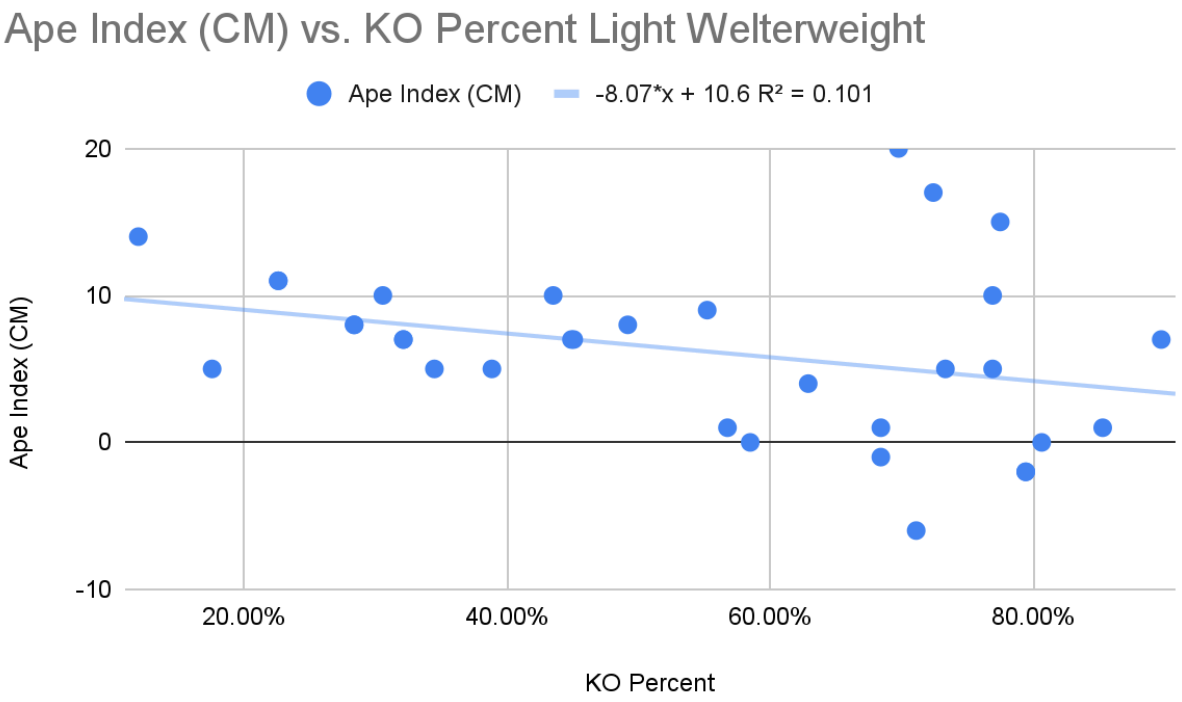
Name	Champion	Ape Index	Z-Score
Pinky Mitchell	0	10	0.65761
Mushy Callahan	1	7	0.12842
Jack 'Kid' Berg	2	5	-0.22438
Tony Canzoneri	3	7	0.12842
Johnny Jadick	4	5	-0.22438
Battling Shaw	5	7	0.12842
Tony Canzoneri	6	7	0.12842
Barney Ross	7	10	0.65761
Tippy Larkin	8	10	0.65761
Carlos Ortiz	9	8	0.30482
Duilio Loi	10	11	0.83401
Eddie Perkins	11	8	0.30482
Duilio Loi	12	11	0.83401
Eddie Perkins	13	8	0.30482
Carlos	14	5	-0.22438

Hernandez			
Sandro Lopopolo	15	5	-0.22438
Takeshi Fuji	16	1	-0.92997
Nicolino Locche	17	14	1.3632
Alfonso Frazer	18	20	2.42159
Antonio Cervantes	19	9	0.48121
Wilfred Benitez	20	0	-1.10637
Aaron Pryor	21	7	0.12842
Julio Cesar Chavez	22	-2	-1.45916
Frankie Randall	23	17	1.8924
Julio Cesar Chavez	24	-2	-1.45916
Oscar De La Hoya	25	5	-0.22438
Kostya Tszyu	26	0	-1.10637
Ricky Hatton	27	-6	-2.16476
Manny Pacquiao	28	4	-0.40078
Danny Garcia	29	1	-0.92997
Terence Crawford	30	15	1.5396
Josh Taylor	31	-1	-1.28277
Teofimo Lopez	32	1	-0.92997

Looking at the data, this does not share the general trend that weight and reach does. Over the years, the ape index does not seem to correlate in either direction. It seems Ape Index is completely random. There are many positive and negative Z scores littered throughout the years, from the first champion Sullivan to our current, Fury. This also does not seem to be dependent on height either. Ape Index is probably something that is not within someone's control, however that makes me wonder if certain fight styles have higher ape indexes on average due to their style requiring it, similar to how basketball players are tall because basketball favors them, and so generally tall people play.

Does Ape Index Determine Power?





Heavyweight Ape Index vs KO Win Percentage

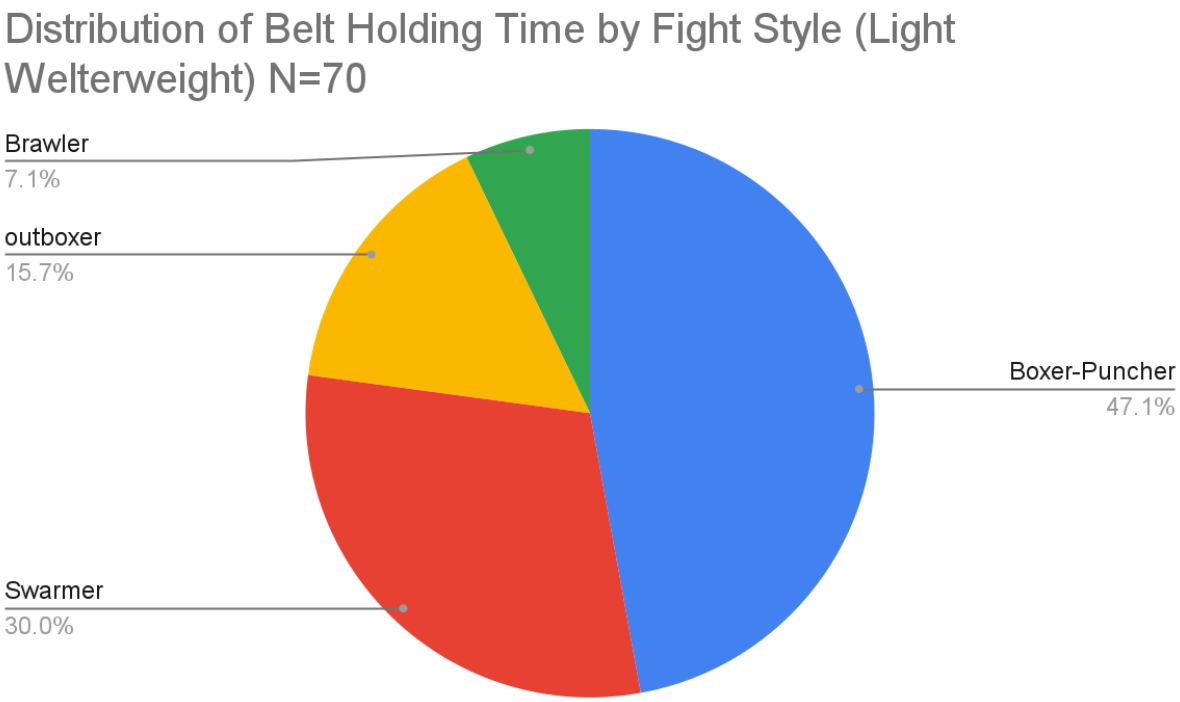
Correlation Coefficient	Coefficient of Determination
r=0.17	r^2=0.03

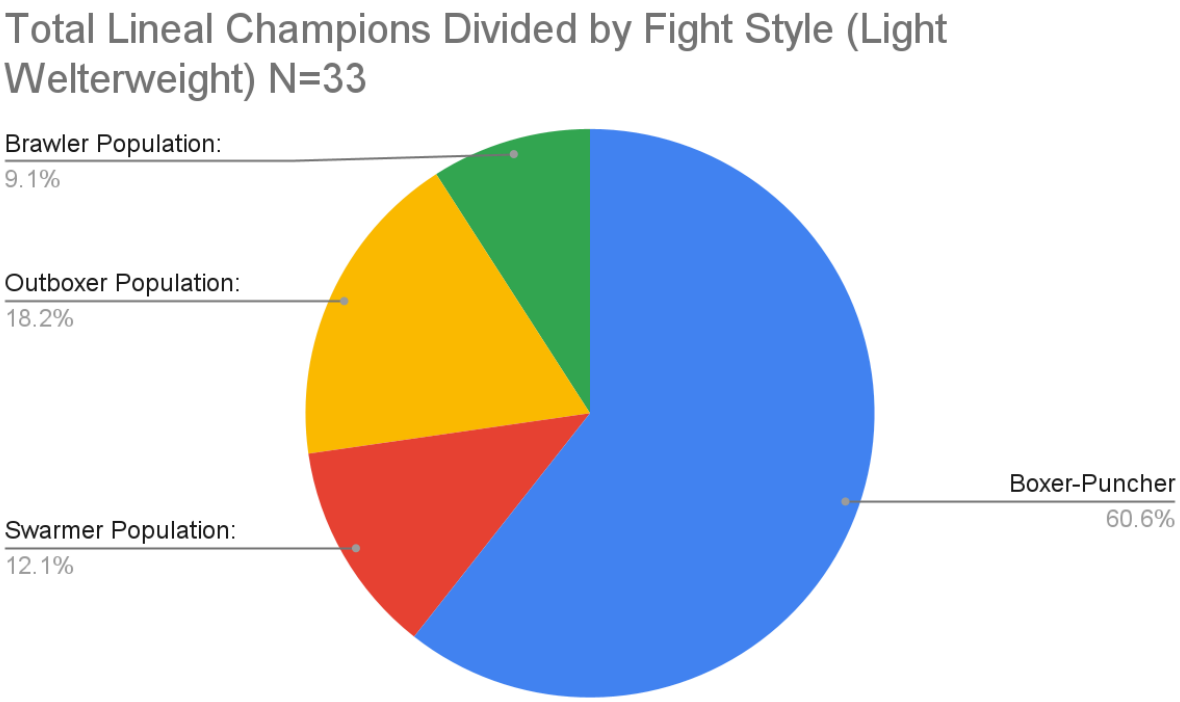
Light Welterweight Ape Index Vs KO Win Percentage

Correlation Coefficient	Coefficient of Determination
0.318	r^2=0.101

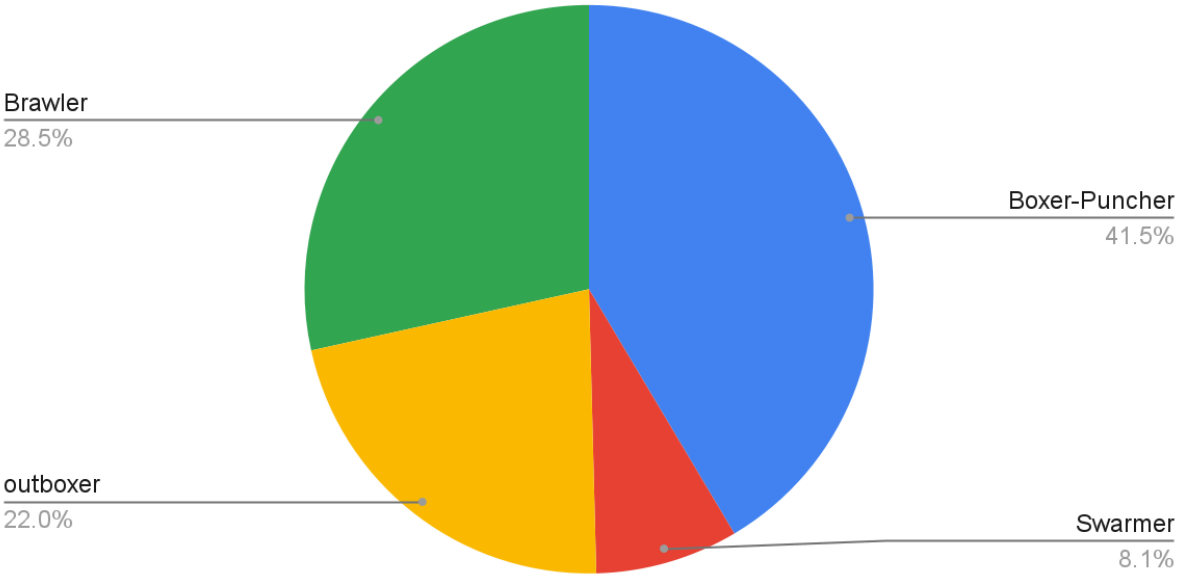
These graphs can also be considered continuous, you can infinitely divide percentages and centimeters.
This went completely against my hypothesis. I had suspected that Ape Index would be strongly positively correlated with KO percentage, but that is not the case.The line of logic I was operating under was the idea that the more length you have in your arms, the harder you would hit, similar to an elastic band being able to generate more force with more material to work with. You could draw the conclusion that ape index is completely irrelevant when it comes to power.
Another important thing to mention is that

What style has been the most Dominant?

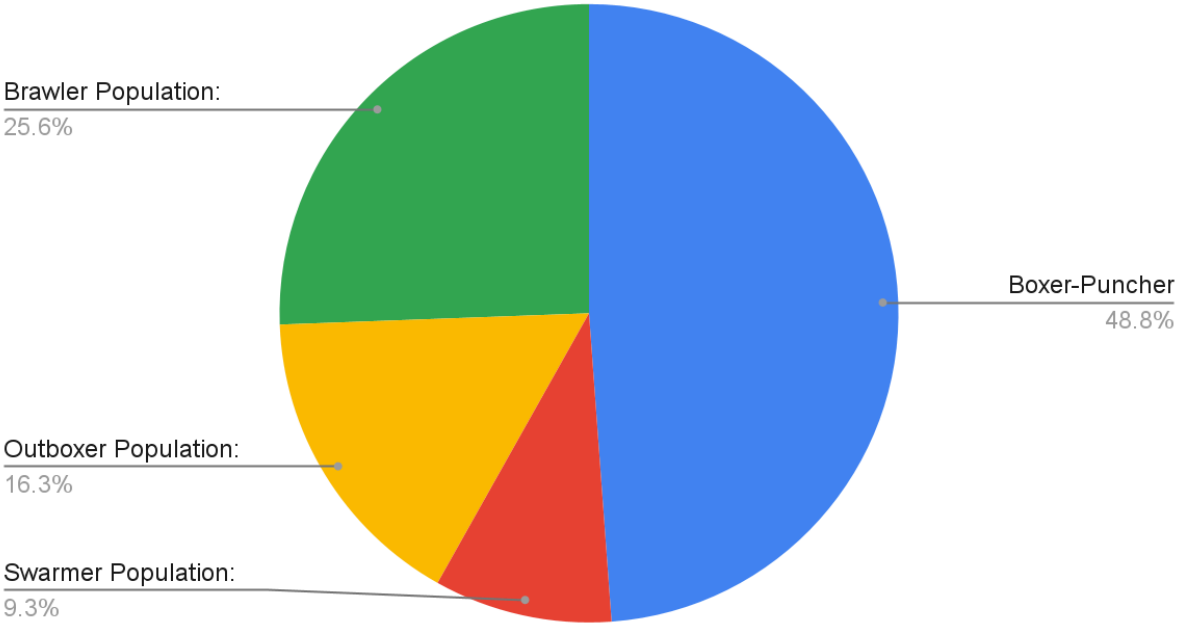




Distribution of Belt Holding Time by Fight Style (Heavyweight) N=127 years



Lineal Champions Divided by Style (Heavyweight) N=43



The above graphs can be described as categorical nominal, as there is no particular order to fight styles. At first, these graphs may show that Boxer-Punchers are the most dominant fighters like how I hypothesized, however if you pay attention to the proportionalities, you can see that they are very evenly distributed. Even among both weight classes, the time held per fight style similarly matches the population of said fight style, with the small exception of the lightweight boxer-punchers, who had a higher population but lower title dominance. This is because there was one boxer in particular who was a swarmer named Tippy Larkin, who held the belt for 11 years. To put that into perspective, the second longest reigning boxer of that weight class held it for 4 years. He single handedly skewed the charts, which is due to the low population size of the lineal champion pool.

Possible Bias'

There is a strong possibility for bias in this data that I have collected, since boxing is not inherently objective, not every fight ends in a KO and so the decision has to be left to the judges, who have had a notorious reputation for corruption and being bribed through mafia connections. To this day, people still argue that corruption plagues the sport, and gives wins to fighters who don't deserve it. This can affect the data because I was specifically researching champions, and if someone made it onto the champion's list without deserving it, they may skew the data with a high amount of wins and low KOs for example. Another bias I have is the lack of criteria I use for these boxers. There's more to boxing than KO percentage, and height and reach. I left out some statistics that could have shined some boxers in a better light which could have changed my results.
Another bias that could skew my results was the low sample size. I collected all lineal champions of the weight divisions, but that's only 30-40 people. One fighter can skew results, which has happened with Tippy Larkin in my "Most Dominant Style" research.
Another bias I have is that I only picked from two websites that are North American, there are other boxing websites that are foreign that I have not considered. This would be a sampling bias.

Conclusion

In conclusion, there is a clear change in boxers in specifically one division: That is, the height and reach of the heavyweight division. Over the years, the heavyweight division has gotten bigger and bigger, and this is because humans as a species have been growing bigger over time as well. The reason why we don't see this phenomenon in other weight classes is because if other weight classes grow in size, they simply move onto the next weight class. Heavyweights don't have to deal with this as they have no other weight class to goto, there is no limit.

Many of my hypotheses were wrong, for one I thought that ape index would have a far greater effect on power than it did. My reasoning was that having a longer arm would be able to generate much more power, but I can see that it is not very relevant. Another thing I did not realize was how balanced the fight styles are. Even though some styles have smaller populations than others, they still are proportionally strong. I thought Boxer-Punchers would have a monopoly due to having no natural counters.

I think the most useful thing to take into account is the idea that weight classes don't matter in how one should box, as we can see from the data the weight classes fight with nearly the same statistics. The ape indexes are nearly the same, and so are the boxing style proportions. I had thought that heavier weights would prioritize power and lower weight classes would prioritize speed, but they seem to go hand in hand.

Appendix

Zeeshan Statistics Project

Sourced Cited

BoxRec. Boxrec. (n.d.). https://boxrec.com/

Compubox. (n.d.). Beta.compuboxdata.com.

https://beta.compuboxdata.com/